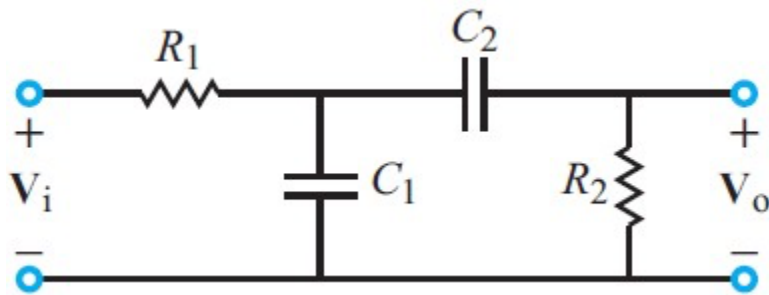


1. (50 pts) For the following circuit, (i) calculate the transfer function $\hat{H}(\omega)$, (ii) Spectral plots for the magnitude and phase of $H(\omega)$ given that $R_1 = 1$, $R_2 = 2$, $C_1 = 1 \mu\text{F}$, and $C_2 = 2 \mu\text{F}$. (iii) The cutoff frequency ω_c and the slope of the magnitude (in dB) when $\omega/\omega_c \ll 1$ and when $\omega/\omega_c \gg 1$.



2. (50 points) For the following circuit, (i) calculate the transfer function $\hat{H}(\omega)$, and (ii) the frequency ω_0 at which the transfer function becomes purely real. (iii) What kind of filter does this circuit implement?

